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Cyber-physical spare parts intralogistics system for aviation MRO

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ABSTRACT

Spare parts management is a vital supporting function in aviation Maintenance, Repair, and Overhaul (MRO). Spare parts intralogistics (SPI), the operational perspective of spare parts management, significantly affects performance of MRO activities. This paper proposes Cyber-Physical Spare Parts Intralogistics System (CPSPIS) to address the synchronization problems associated with the SPI business process and SPI resources. The proposed system applies Internet-of-Things technologies and unified representations to provide resources and operations traceability and visibility. Further, CPSPIS contributes several services with self-X abilities for real-time synchronization throughout the SPI process. The communication service in the physical layer provides a self-adaptive ability to coordinate between the physical environment and cyberspace. The tracking and tracing service uses self-configured SPI business-related functions to capture accurate real-time resource and operation data through spatial modeling and location-based components for system resource coordination. The self-organized data source integration service consolidates all SPI request sources as the starting point for process analysis. The initialization service has self-awareness capability and models the transformation between the request source and SPI operations. The self-adjusted execution service synchronizes every two operation steps using real-time resource and operation data. In addition, CPSPIS develops applications and visualization tools for real-time cooperation between execution and decision-making. A real-life case study is conducted in one of the largest aviation MRO organizations, and the quantitative and qualitative improvements of CPSPIS are discussed.

1. Introduction

Spare parts management (SPM) is fundamental in providing aircraft spare parts to support the activities of aviation Maintenance, Repair and Overhaul (MRO) [1]. SPM performance significantly affects the performance of maintenance activities. Untimely delivery of spare parts for aircraft-on-ground (AOG) demand incurs high aircraft maintenance costs and causes flight delays. SPM for aviation MRO providers usually includes spare parts for different aircraft types for multiple airlines. With application of spare parts pooling strategies in recent years, SPM faces more significant challenges in complex spare parts intralogistics (SPI). Recent studies have proposed several solutions from managerial and system perspectives to address complex SPI. Product lifecycle management (PLM) is implemented with a focus on maintenance planning and scheduling, and supports spare parts-driven process tracking and tracing using only intralogistics execution results [2]. Some information systems have tried to bridge the information gap, including Enterprise Resource Planning (ERP) systems and the next-gen Integrated Facility Platform (IFP) [3]. PLM and information systems provide the essential SPM; however, SPI operation management still faces several challenges in MRO organizations.

The first challenge is the complex collaboration of multiple participants at the execution level. SPI processes thousands of spare parts handling requests in daily operations from logistics customers and maintenance jobs [1]. The handling process for these requests usually involves at least three steps from several working teams, implementing more than twenty types of activities. Cooperation between participants remains challenging in the dynamic and heavy SPI process. Participant decision-making also faces great uncertainty due to frequent requirement changes during handling of spare parts. For example, spare part delivery destination and handling priority can be amended at any time before being issued to the maintenance team. Emergency cases such as aircraft-on-ground (AOG) demand and ad hoc customer shipping requests require high-frequency real-time information sharing between SPI working teams.

The second challenge is the strict lifecycle tracing and tracking of the

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entire spare parts-driven process. New spare parts are tracked from receipt to installation in an aircraft, including quality inspection and other turnaround operations in SPI. All used non-consumable parts follow the gain-one-return-one rule, with tracing of disassembly from an aircraft, SPI operations, and destruction during the return process. Handling of new and returned parts is documented with operation records that include handler signatures and paper certifications. Handling of normal cases and exceptions during the SPI process is essential for the quality check to ensure the MRO safety operation specifications.

The third challenge is operation and resource coordination in a short timeframe. Handling time is shortened (minutes or hours) for urgent part requests for unplanned maintenance jobs. When a complex handling process is urgent, coordinating the spare parts, the required equipment, and the available workers becomes challenging for SPI operation management. For instance, inspection, warehousing operations, delivery, and an airport security check should be completed within one hour for AOG demand to serve an aircraft in operation. In contrast, the time to complete scheduled maintenance jobs is two to twenty-four hours. Scheduled jobs require many spare parts; handling of these parts is limited by the capacity of human resources and vehicles. For example, hangar maintenance jobs require preparation of dozens of parts with different handling processes, within two hours.

Cyber-physical Systems (CPS), one of the most advanced information systems, are helpful in addressing these challenges. CPS facilitate traceability and visibility of product logistics resources for monitoring machine and shopfloor [4,5]. CPS are widely used in smart manufacturing to increase the interconnectivity of the physical environment and cyberspace [6]. However, machine-oriented CPS are unsuitable for SPI in aviation MRO. First, data acquisition for the SPI business process requires a specific design, as spare parts come in all types, shapes and sizes and cannot be manufactured with sensors like machines. Second, business synchronization revolves around mobile spare parts instead of planned product orders. The circulation of spare parts often faces changing destinations and urgent demands, especially for unplanned maintenance jobs. Third, compared to the stable configuration of machines and shopfloors in manufacturing, SPI faces more uncertainty at the execution level. CPS for SPI retains greater flexibility for handling spare parts from different participants. Although attempts have been made to introduce radio-frequency identification (RFID) technologies for aviation SPI, an overall CPS-enabled solution considering these challenges requires further investigation.

This study proposes a CPS-enabled solution to address these challenges. Three research questions are presented:

1 How to synchronize the business process with uncertainty and flexibility in operations so that SPI can facilitate a fast response to requests?
2 How to abstract and encapsulate physical resources such as human resources, machines, and spare parts to facilitate traceability and visibility?
3 How to design a mechanism to facilitate synchronization between the business process and resources in a short time window?

A cyber–physical spare parts intralogistics system (CPSPIS) is proposed to address these problems. A system architecture based on CPS is presented as the overall solution. The Internet-of-Things (IoT) paradigm empowers all resources with intelligence to capture real-time resource data including human resources, equipment, logistics units, and spare parts. A spare parts-oriented business process is systematically established using unified representations to capture real-time operational data. Several self-X mechanisms (self-adaptive, self-organized, self-aware, self-adjusted, and self-configured) are proposed for resource coordination and process synchronization through embedded real-time resource and operation data. Initialization and execution services are used to model SPI execution and decision-making. The proposed system is implemented in a real-life case to evaluate its performance and benefits.

The remainder of this paper is organized as follows. Section 2 reviews related research in aviation SPM and associated technologies. Section 3 introduces the system architecture of the proposed CPSPIS. Section 4 highlights the self-X services in CPSPIS. A case study is conducted, and its implementation and improvements are discussed in Section 5. Section 6 presents the conclusions.

2. Related work

2.1. Spare parts management in aviation MRO

Many studies have addressed SPI challenges by designing spare parts inventory systems; this study considers SPI operation management [7–9]. The complexity of SPI is the result of spare parts logistics and internal maintenance scheduling in four areas. First, SPI is sensitive to downtime costs in preparing high-value, user-specific parts based on critical and specific control strategies [10]. Second, participants collaborate more intimately and fulfill time-sensitive customer requests using a parts pooling strategy between aviation MRO organizations [11]. Third, SPI measures all performance in time-phased services, such as just-in-time parts delivery and parts receipt for scheduled maintenance jobs [1]. Fourth, SPI produces frequent ad hoc activities to respond to low and irregular demand [10]. The complexity of SPI can be reduced by enhancing information sharing between participants, improving operation and resource tracking and tracing, and shortening the time required for operation and resource coordination.

Several attempts have been made using RFID to improve the traceability and visibility of the SPI process to reduce complexity. Ngai et al. built an analytical model to examine and evaluate RFID values in parts inventory monitoring [12]. Chang et al. developed an RFID-enabled maintenance system with RFID implementation to enable visibility of parts [13]. For operation monitoring, experiences and lessons from implementing RFID in SPI have been summarized from an actual case company [14]. Saygin and Tamma applied RFID for shared resource tracking, including tools, equipment, and materials during field maintenance [15]. IATA lists RFID as a key enabling technology in its further system development direction known as Digital Aircraft Operations [16]. Although the value of RFID in spare parts tracking and tracing has been widely discussed, this study explores how RFID and other advanced technologies can be implemented in a real-life case.

Previous use of RFID for SPI data collection had two limitations. First, these applications tracked spare parts but lost the perspective of other resources such as workers and vehicles, producing difficulty in analyzing the complex SPI process. Second, these applications were limited to single early IoT technology, like RFID, with limited data collection capability. A wide range of newly developed IoT technologies, such as tags, sensors and mobile industrial devices, can be integrated to enhance SPI data analysis in terms of scale and quality. For example, environmental data can be collected automatically using Bluetooth sensors to monitor storage condition. IoT technologies and their benefit in managing SPI resources require further study [17]. The research gap in the literature is in an overall solution for tracking and tracing spare parts, operations, and other resources in the entire process based on IoT technologies.

2.2. CPS and IoT technology and applications

CPS and IoT are similar concepts that emerged almost simultaneously to address critical issues in designing next-generation computing systems. Their similarities are in applying and bridging heterogeneous technologies such as RFID, tags, sensors, actuators, and mobile devices for synthesizing physical and cyber capabilities. As the next generation of systems, CPS focuses on integrating knowledge and engineering methods to design new computing mechanisms and supporting technologies [18,19]. The IoT paradigm aims to develop various technologies to increase intelligent physical data collection including identification and sensing, gateway, communication and network [20]. Many essential standards for these IoT technologies are also established in terms of perception, transmission, computation and application layers [21]. Early studies in manufacturing and logistics industries used RFID technologies for resource and operations tracking [22,23]. Service-oriented architecture (SOA) is a typical hierarchical structure for implementing IoT infrastructures with computing ability [24,25]. From early RFID to the concepts of CPS and IoT, many studies have discussed system architectures, enabling technologies and domain-specific applications in the last decade. However, the mechanisms that integrate the enabling technologies and domain-specific knowledge need further discussion.

5C-level CPS is proposed with connection, conversation, cyber, cognition, and configuration capabilities for manufacturing [6]. Enabling technologies such as near-field communication, wireless sensor networks, and their relevant protocols have been widely discussed [24,26]. Research fields such as healthcare, manufacturing, food supply chain, production logistics and modular integrated construction benefit from use of IoT and CPS for tracking and tracing [27–30]. In recent years, CPS-based and IoT-enabled applications have connected domain-specific applications and intelligent decision-making ability. In manufacturing, [5] and [31] proposed advanced analytical methods to construct intelligent shop floors with enhanced adaptive capability [5,31]. Morth et al. and Lee et al. presented CPS-based applications for operation management and performance monitoring using data analysis techniques in the logistics field [32,33]. Although researchers have deeply analyzed the potential benefits of CPS and IoT in manufacturing and logistics, few studies have considered applications in maintenance logistics. To address the research gap, a novel architecture based on five-layer CPS, IoT technologies, and data visualization techniques is proposed in this study for spare parts management in the aviation MRO industry.

3. Cyber-physical spare parts intralogistics system (CPSPIS)

This section introduces CPSPIS to address SPI synchronization problems. Fig. 1 shows the overall CPSPIS system architecture following the typical concept of CPS. Physical space provides a preliminary realization of resource coordination by constructing the smart SPI working environment and making the SPI resources visible. Physical space uses IoT technologies for the SPI working area, workers, equipment, logistics units, and spare parts in the physical object layer (POL), and models these SPI resources as digital objects in the smart object layer (SOL). Further, it delivers object fusion and communication service (OFCS) with self-adaptive ability to the SPI working environment by designing object conversion and communication models. The gateway layer (GL) in cyber space uses two middleware systems capable of object digitalization for real-time data transactions between the cyber and physical environments. The service layer (SL) contributes to the synchronized business process. Requests, tasks, workflow, and the corresponding pools manage the SPI process for linking the request source, execution steps, and historical tracing for flexibility. Heterogeneous data source integration service (HDSIS), EFSM-enabled spatial–temporal initialization service (ESTIS), and EFSM-enabled spatial–temporal execution service (ESTES) constitute the dynamic mapping mechanism between

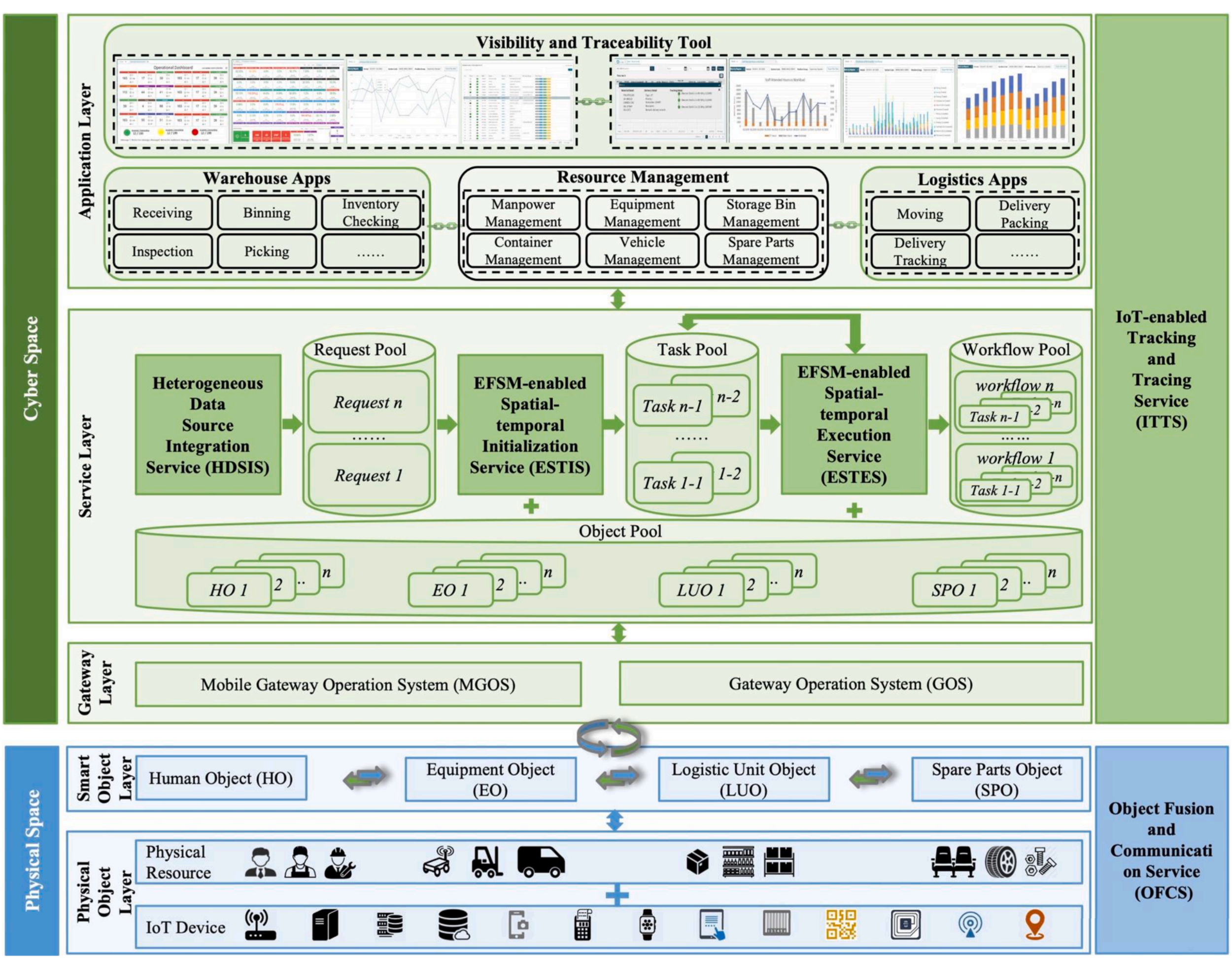

Fig. 1. CPSPIS system architecture.

resource and operation steps for SPI execution and decision-making using the digital representation from the object pool and the real-time spatial–temporal data from the IoT-enabled tracking and tracing service (ITTS). HDSIS manages various heterogeneous business process requirements for CPSPIS compatibility and interoperability with existing systems and manual instructions. ESTIS models the transformation between request and task with self-awareness; ESTES models the conversion of tasks with self-adjustment capability. ITTS automatically realizes real-time spatial–temporal traceability and visibility of resource and location-based task dispatching for front-end execution. The application layer (AL) visualizes the synchronized results for user interaction, including the execution, monitoring, tracing, configuration, and spatial–temporal analysis of the business process and the resource.

3.1. Physical space

3.1.1. Physical object layer (POL)

The POL defines two types of physical objects, IoT devices and physical resources. IoT technologies include Bluetooth beacons, active and passive RFID tags, wearable devices, mobile phones, smart gateways, smart tags, mobile printers, and smart sensors. These IoT devices make physical resources visible with unique identifications and provide detection ability, communication ability, and edge computation ability in the physical environment of SPI. The POL includes four essential physical resource types: human, equipment, logistics units, and spare parts. With the definition of POL, the physical space of CPSPIS can integrate all required resources in all SPI operations. Equation (1) uses the physical object *po* to represent part of a resource or IoT device, and constructs the physical object set PO by combining physical resource set PR and IoT device set DEV.

$$PO = PR \cap DEV, po \in PO \tag{1}$$

The POL facilitates the intelligent physical environment to achieve three objectives: 1) to construct the basis of real-time data collection with the IoT device and physical resource, 2) to make all types of physical resources visible and traceable in CPSPIS, 3) to produce physical resource intelligence for communication and connection.

3.1.2. Smart object layer (SOL)

The SOL provides a unified format for four heterogeneous resources so that they can be further operated in CPSPIS cyberspace. A smart object is a pair of heterogeneous physical objects, including part of a physical resource and an IoT device. The SOL introduces human objects (HO), equipment objects (EO), logistics unit objects (LUO), and spare parts objects (SPO), which have a one-to-one correspondence with the four types of physical resources in the POL. For example, LUO can be storage units (bin, shelf) or transfer units (boxes, pallets, cages, mobile workstations) equipped with smart sensors in SPI. Equation (2) formulates smart object set SO. Equation (3) defines *so* as the pair $\langle pr, dev \rangle$, and provides flexibility for smart-object aggregation and decomposition with easy replacement of physical resources or IoT devices.

$$SO = PR \times DEV, PR \subset PO, DEV \subset PO \tag{2}$$

$$so = \langle pr, dev \rangle, pr \in PR, dev \in DEV, so \in SO \tag{3}$$

3.2. Cyber space

3.2.1. Gateway layer (GL)

The GL transfers smart objects from the SOL to the service layer (SL) to connect cyber- and physical space. Previous studies by two co-authors contributed to the design of the gateway system, including the Mobile Gateway Operation System (MGOS) and Gateway Operating System (GOS) [34–37]. The GOS included an easy-to-deploy and simple-to-use RFID system to support object identification in cloud-based scenarios. The MGOS included a lightweight, reconfigurable, plug-and-play system to support mobile operation scenarios. The GL uses MGOS and GOS for registration, publishing, and searching of smart objects including human objects, equipment objects, logistics unit objects, and spare parts objects for the complex SPI working environment.

3.2.2. Service layer (SL)

The SL achieves three objectives: 1) to provide a unified resource and process definition of SPI to allow a configurable CPSPIS workflow; 2) to enable real-time mapping of physical resources and each activity in the process to allow CPSPIS to address the synchronization issue between resources and business processes from time to time; 3) to support mobility in the operational level and flexibility in the decision-making level for SPI. Fig. 1 shows the working logic of four pools and three services in the SL. With interaction from the GL, the object pool represents the SPI resource chain at the bottom of the SL. HDSIS, the starting point of the business process chain in SPI, integrates different SPI request sources. ESTIS and ESTES provide the capability for task initialization and task execution, respectively. The request pool, task pool, and workflow pool act as the input and output between two services. Requests manage all SPI request sources to organize the business process. Instead of managing a complex workflow case by case for each request, CPSPIS deconstructs requests into tasks to increase operation flexibility. Tasks are released to front-end operators in real-time, making daily execution transparent at the operational level. The workflow arranges the requests and executed tasks in chronological order, allowing accurate tracking of the spare parts-driven process and historical tracing with allocated resource details. The four pools contribute the output to the AL for data visualization. HDSIS, ESTIS, and ESTES are further explained in Section 4. The formulations of the four pools and their elements are described as follows.

- Object pool

The object pool directly obtains smart objects from the GL and combines them with real-time information from ITTS. In the SL, the object pool constructs an n-dimensional object space for ESTES and outputs the digitalization of physical objects for the AL. The objects in the object pool are the digital instantiations of smart objects, extended with real-time spatial–temporal information from ITTS. Equations (4) and (5) provide simple definitions for object pool *O* and its element *o*. In Equation (4), *SO*, *PA*, *SA*, and *TA* represent smart objects, physical attributes, spatial attributes, and temporal attributes, respectively, to integrate real-time spatial–temporal data from IoT-enabled services. In Equation (5), each element *o* is an ordered pair, representing the smart object and its corresponding real-time information.

$$O = SO \times PA \times SA \times TA \tag{4}$$

$$o = \langle so, pa, sa, ta \rangle, pa \in PA, sa \in SA, ta \in TA, so \in SO, o \in O \tag{5}$$

- Request pool

The request pool receives the abstract definition from HDSIS to generate and store the instantiation of one specific request according to its request type. The requests in the request pool are the input for ESTIS, and the source data for the traceability and visibility tool in the AL. The request *r* and its relationship with request pool R is

$$r \in R \tag{6}$$

- Task pool

The task pool serves as the connection between ESTIS and ESTES, as shown in Fig. 1. Tasks are elements in the task pool, representing the minimum SPI operation step. Task definitions cover the normal, exceptional, and error operation steps that form a workflow. Tasks reduce the duplicate workflow definition to make the business process clear, flexible, and configurable. For example, CPSPIS splits a request to receive spare parts in SPI into four tasks, including package receipt, part receipt, part check, and part inspection. CPSPIS can automatically allocate different operators with different skills and tools in different workstations for each task. Task pool T and its element t are defined in Equations (7) and (8).

$$T = R \times SPO \times HO \times EO \times LUO, SPO \subset O, HO \subset O, EO \subset O, LUO \subset O \tag{7}$$

$$t = \langle r, spo, ho, eo, luo \rangle, r \in R, spo \in SPO, ho \in HO, eo \in EO, luo \in LUO, t \in T \tag{8}$$

- Workflow pool

The workflow pool stores the output transitions from ESTES and provides the basic definition and execution result for the AL. Equations (9) and (10) are the definitions of workflow, which is an element in the workflow pool. The formulation of workflow is a dynamic concatenation of several tasks performed in sequence, and links the completed tasks as nodes of itself. The workflow stores task information with the corresponding physical resource and the transitions between operational steps throughout the SPI business process.

$$W = T \times T \tag{9}$$

$$w = \langle t_i, t_j \rangle, t_i, t_j \in T \tag{10}$$

3.2.3. Application layer (AL)

The AL allows user-friendly interaction for SPI participants such as operators, supervisors, and managers. The physical resources and the business process are synchronized throughout the planning, execution, and evaluation stages.

- Warehouse apps and logistics apps offer front-end and web-based applications at the operational level. Operators and supervisors access the authorized functions to respond to their tasks. They record the operation errors and exception cases in a mobile and flexible manner.
- Resource management (RM) refers to management of all resources in CPSPIS. In planning and scheduling, managers configure and coordinate spare parts, equipment, logistics units, and workers in different workstations. Supervisors monitor real-time status and allocate resources during the execution stage.
- The visibility and traceability tool (VTT) tracks resources and traces the business process through data analysis and visualization techniques. First, the VTT accurately keeps track of all activities in SPI. After receiving the request, task, and workflow from the SL, the VTT models them with spatial–temporal information in different colors. Real-time data can be monitored and distribution data in different spaces can be analyzed from a general perspective. Second, the VTT evaluates performance throughout the SPI process. The normal cases and exceptions during SPI process are chained clearly for tracking. The historical data is visualized and structured into dashboards. The dashboards show the statistics in terms of resources, including tasks, workflows, and requests. The dashboards provide analysis for synchronization suggestions before execution (workflows). System users can intuitively access these statistics and comparison data to optimize their decision-making.

4. Self-X services in CPSPIS

CPSPIS delivers five services including OFCS, HDSIS, ESTIS, ESTES, and ITTS to achieve self-adaptive, self-organized, self-aware, self-adjusted, and self-configured CPS features [38].

4.1. OFCS

OFCS integrates the POL and SOL in physical space and addresses two technical issues. First, OFCS designs smart object aggregation and decomposition (SOAD) for conversion between physical objects and smart objects, as shown in Fig. 2(a). Two heterogeneous physical objects, such as part of a physical resource and an IoT device, can be combined into a smart object; two homogeneous objects cannot be aggregated. A smart object can be flexibly disassembled into two physical objects, allowing replacement of either. CPSPIS adapts to changing resource conditions through SOAD. Second, OFCS uses a smart object communication (SOC) mechanism for information interaction between any two smart objects, as shown in Fig. 2(b). SOC models real-world operating scenarios, such as a worker picking spare parts or using a vehicle, so that cyberspace is adaptive to the interactions in physical space over time.

Both SOAD and SOC embed OOM and simple FSM. OOM offers encapsulation, inheritance, and polymorphism to define heterogeneous elements in a unified representation [39]. Simple FSM, a famous computational model, can simplify the definition of state transaction [40–43]. SOAD is defined as a six-tuple $(\mathrm{I}, \mathrm{O}, \mathrm{S}, s_0, T, F)$, where

I is a finite non-empty set of inputs, $I = \{PO, SO\}$.

P is a finite non-empty set of outputs, $P = \{PO, SO\}$.

S is a finite non-empty set of states, $S = \{PO, SO\}$.

s_0 is the initial state, an element of S.

T is the state-transition function, $T : S \times I \rightarrow S \times P$.

F is the output function, $F : S \times I \rightarrow P$.

PO and SO are the elements in the POL and SOL, respectively. There are four transactions in this stage:

$T1 : (\mathrm{PO}, \mathrm{PO}) \rightarrow (\mathrm{SO}, \mathrm{SO})$: two heterogeneous POs are aggregated into an SO.

$T2 : (\mathrm{PO}, \mathrm{PO}) \rightarrow (\mathrm{PO}, \mathrm{PO})$: two homogeneous POs are output.

$T3 : (\mathrm{SO}, \mathrm{PO}) \rightarrow (\mathrm{SO}, \mathrm{SO})$: SO is output when the input PO is different from any of the pair <pr, dev> of SOs.

$T4 : (\mathrm{SO}, \mathrm{PO}) \rightarrow (\mathrm{PO}, \mathrm{PO})$: SO is decomposed into POs when the input PO is the same as any of the pair $< pr, dev >$ of SOs.

In addition to PO and SO, a gateway object (GO) is defined to represent the elements in the GL. In Fig. 2(b), SOC is a six-tuple $(\Sigma, \Gamma, \mathrm{C}, c_0, \delta, \omega)$, where

Σ is a finite non-empty set of inputs, $\Sigma = \{ < o_1, o_2 >, \{o_1, o_2\} \}$.

Γ is a finite non-empty set of outputs, $\Gamma = \{ < o_1, o_2 >, \{o_1, o_2\} \}$.

C is a finite non-empty set of states, $\mathrm{C} = \{connected, disconnected\}$.

c_0 is the initial state, an element of C.

δ is the state-transition function, $\delta : C \times \Sigma \rightarrow \mathrm{C} \times \Gamma$.

ω is the output function, $\omega : C \times \Sigma \rightarrow \Gamma$.

In the definition, $\{o_1, o_2\}$ is a two-element set where $o_1, o_2 \in \{SO, GO\}$, and $< o_1, o_2 >$ is an object pair where $o_1, o_2 \in \{SO, GO\}$. The transactions in this machine can be categorized as two types. $\delta_1 : C \times \Sigma \rightarrow \{\mathrm{disconnected}\} \times \{o_1, o_2\}$ indicates that when the next state is *disconnected*, the output is the two-element set $\{o_1, o_2\}$. $\delta_2; (C \times \Sigma \rightarrow \{\mathrm{connected}\} \times < o_1, o_2 >)$ indicates that when the next state is *connected*, the output is the object pair $< o_1, o_2 >$.

4.2. HDSIS

HDSIS is the start of the SPI process chain in the SL to provide interoperability. SPI handles requests from planning and scheduling of spare parts management and maintenance jobs, as shown in Fig. 1. The possible request sources are existing systems, manual creation in the AL, and auto-triggered SPI internal requests from system schedulers. HDSIS organizes these heterogeneous request instructions to model the process chain through RESTful APIs. After processing all data from RESTful APIs, HDSIS extracts valuable information such as request name, request

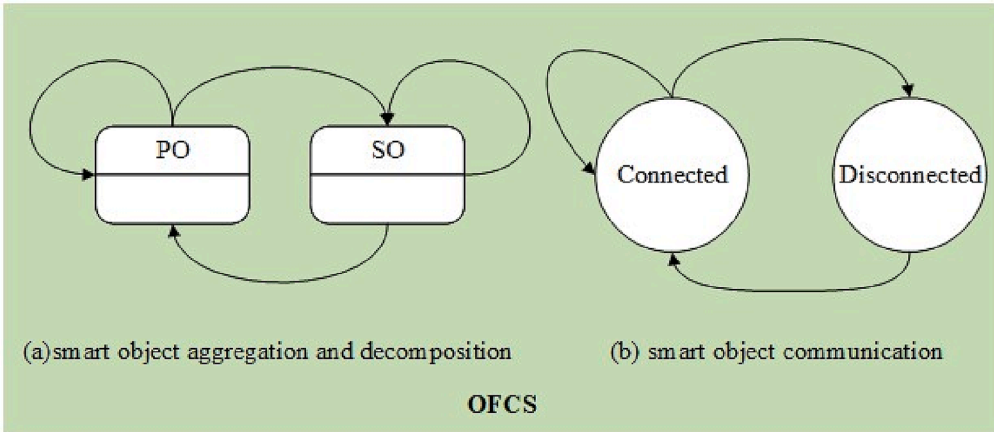

Fig. 2. Object fusion and communication service (OFCS).

priority, required resources, and other information. It structures the information into a unified representation so that HDSIS is compatible with requests in different formats. The working logic of HDSIS is presented in Fig. 3.

4.3. ESTIS

ESTIS transforms requests into tasks with self-awareness capability. ESTIS is aware of the available real-time resources. Receiving real-time spatial–temporal resource information from ITTS and the object pool, ESTIS presents the capacity and availability of spare parts, workers, logistics units, and equipment for handling requests. ESTIS is aware of the real-time request data. Derived from the request data in the request pool and the configuration of feasible request types from the AL, ESTIS systematically organizes all workflows in SPI. ESTIS is aware of the first operation tasks for all received requests. ESTIS automatically incorporates the request data, configuration data, and resource data into the EFSM-enabled spatial–temporal initialization model (ESTIM) to decide the type of task to be triggered. ESTIS initializes the first operation task for each request, adapted to available spare parts, idle workers, and available logistics units. ESTIS triggers different first tasks for a request type using the multi-dimension spatial–temporal resource information to synchronize the operations in the SPI process and resources in real time. Fig. 4(a) shows the working logic of ESTIS; Fig. 4(b) presents a diagram of ESTIM.

The ESTIM model is defined as a seven-tuple $\{IS, II, IO, ID, IF, IU, IT\}$, where

IS is a set of request states; states include start (s) and end (e).

II is a set of requests indicating the real-time request data from the request pool.

IO is a set of tasks, indicating the first task for each input request. The output task is assigned with real-time object data and matches the first task type from the rule-based configuration.

ID is a four-dimensional IoT-enabled real-time object data space, $ID_{spo} \times ID_{ho} \times ID_{eo} \times ID_{luo}$. We define x as a four-dimensional vector with components $x_i \in ID_i$; x_i is a three-dimensional space; $D_{basic} \times D_{spatial} \times D_{temporal}$. D_{basic} indicates the fundamental attributes of an object; $D_{spatial}$ describes the spatial attributes of an object; $D_{temperal}$ describes the temporal attributes of an object;$d_i \in x_i$ represents a full piece of object information.

IF is a set of enabling functions that indicates the working physical and cyber environments of CPSPIS, $if_i : ID \rightarrow \{0, 1\}$.

IU is a set of updating functions iu_i such that $iu_i : ID \rightarrow ID$. NOP is an updating function when CPSPIS receives an abnormal request. Updating object attributes is another updating function for regular requests.

IT is a transition relation that explains the type of first-step task triggered from the input request based on the real-time data, $IT : IS \times IF \times II \rightarrow IS \times IU \times IO$.

Further, any pair $< is, id >$, $is \in IS \text{ and } id \in ID$ is known as a configuration, which indicates any type of object at the start state or end state. As only one transaction occurs once the real-time object and real-time request are obtained, ESTIM is stable.

4.4. ESTES

ESTES addresses task conversion during the SPI process. Fig. 5(a) shows the working logic of ESTES. ESTES obtains the real-time operation data and rule-based configuration from the AL, task data from the task pool, and spatial–temporal resource data from ITTS and the object pool. The EFSM-enabled spatial–temporal execution model (ESTEM) models the execution using these three data sources and the rule-based configuration. Task and workflow are the results of ESTEM. ESTEM is defined as a seven-tuple $\{ES, EI, EO, ED, EF, EU, ET\}$, where

ES is a set of tasks defined in the task pool.

EI is a set of real-time data from all types of objects.

EO is a set of workflows; eo_i is a sequenced list $\{< es_1, ei_1 >, \cdots, < es_i, ei_i >\}$ that indicates the sequence of tasks performed for a request. Each

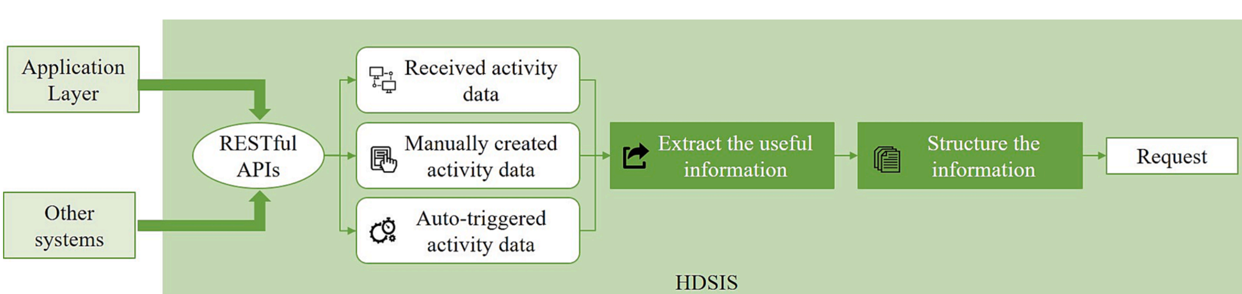

Fig. 3. Heterogeneous data source integration service (HDSIS).

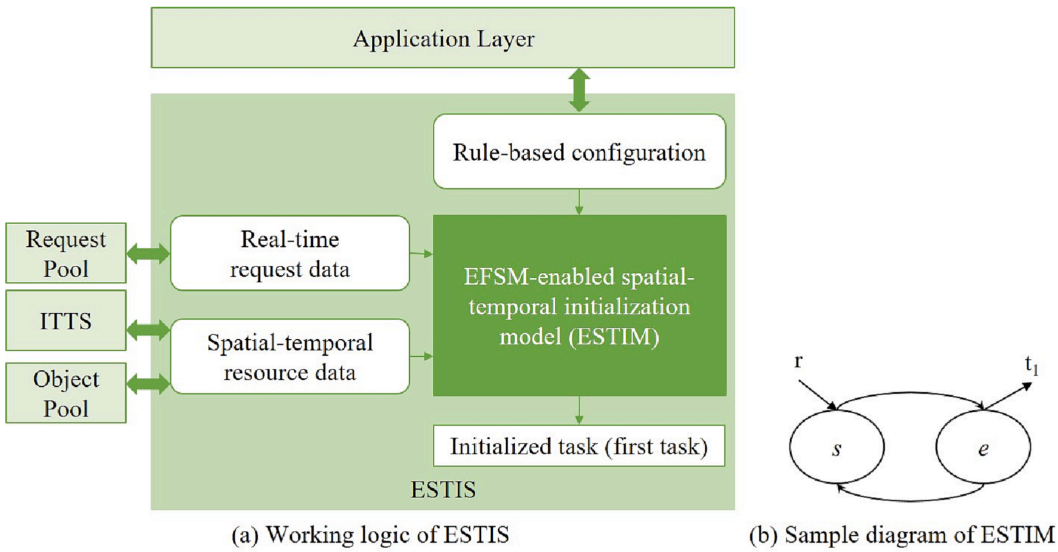

Fig. 4. EFSM-enabled spatial–temporal initialization service (ESTIS).

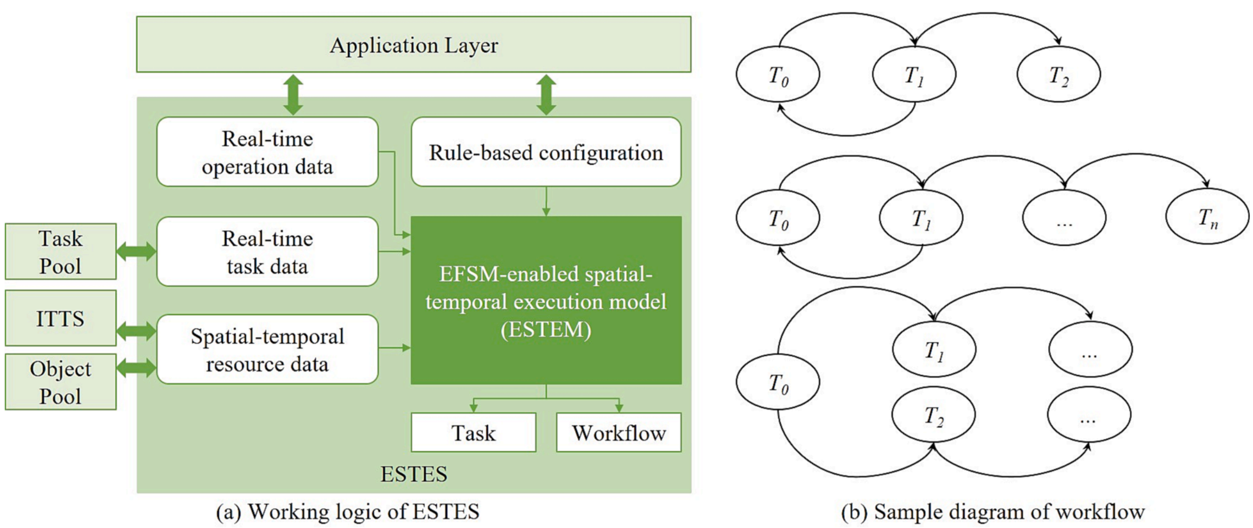

Fig. 5. EFSM-enabled spatial–temporal execution service (ESTES).

pair $< es_i, ei_i >$, $es_i \in ESandei_i \in EI$, indicates the performed task with the corresponding resources for operation step i. The initial pair $< es_1, ei_1 >$ is decided by ESTIS.

ED is a 4-dimensional IoT-enabled real-time object data space, $ED_{spo} \times ED_{ho} \times ED_{eo} \times ED_{luo}$. We define y as a four-dimensional vector with components $y_i \in ED_i$. y_i is a three-dimensional space, $E_{basic} \times E_{spatial} \times E_{temporal}$. E_{basic} indicates the fundamental attributes of an object; $E_{spatial}$ describes the spatial attributes of an object, and $E_{temperal}$ illustrates the temporal attributes of an object.$e_i \in y_i$ represents a full piece of object information.

EF is a set of enabling functions that indicates both the request information and the required resources for the first task of the request.

EU is a set of updating functions that updates the status and other information for the objects in the object pool.

ET is a transition relation that explains how a request transfers from one task to the next task using the real-time data, $ET : ES \times EF \times EI \rightarrow ES \times EU \times EO$.

ESTES dynamically generates a task for real-time tracking and a workflow for tracing for each request. Fig. 5(b) displays three possible scenarios for one request. ESTES achieves self-adjustment for execution of SPI. First, ESTES triggers tasks by dynamically matching the real-time spatial–temporal resource information. Second, ESTES automatically triggers the current task. In ESTEM, the current task is decided by completing the previous task and real-time resource information. The connection and transition between two tasks represent how the operation steps are performed in SPI. Third, ESTES chains the completed tasks to adjust to real-time operational data to form the workflow. The workflow for each request is generated up to the current task. The executed tasks of each workflow are recorded with the resource information, including who performed the task and which machine was used.

4.5. ITTS

Fig. 6 illustrates the technical structure of ITTS. ITTS provides an authority engine, data synchronization engine, spatial modeling engine, and location-based engine for spatial–temporal traceability and visibility of the resources and business process. The authority engine provides access control for the system by defining the privilege, protocol, message, and device. The data synchronization engine manages data transaction with computing infrastructures, including data formatting, storage, processing, and backup. The spatial modeling engine and location-based engine provide the business-related functions for SPI.

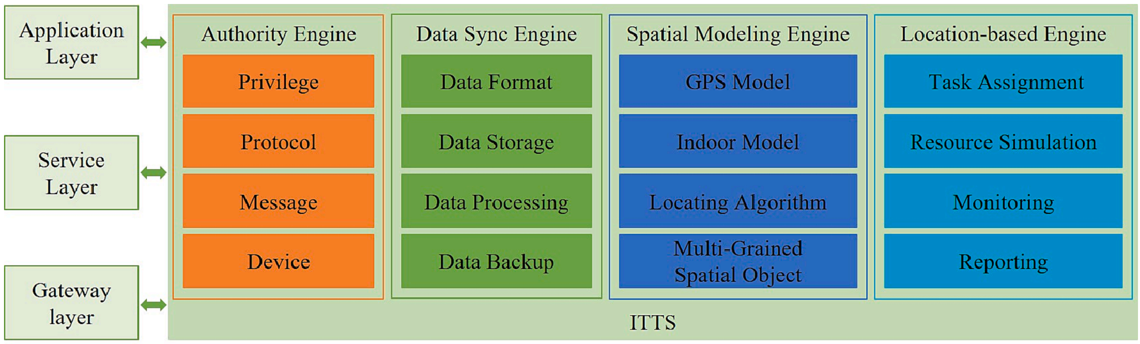

Fig. 6. IoT-enabled tracking and tracing service (ITTS).

The spatial modeling engine applies a GPS model, an indoor model, and locating algorithms to obtain indoor and outdoor environment information. Different spaces in SPI, such as storage containers, shelves, trolleys, stores, delivery points inside the hangars, and repair stations are transformed into multi-grained spatial objects that configure their resource capacity and available resource information in real time using the constantly changing object data from the GL. The self-configured resource distribution in different spaces is accurate and visible for resource allocation, reducing the coordination time. The spatial–temporal resource information provided by the spatial modeling engine is available for synchronization in the SL. Based on the mapping of resource data and task data into spatial objects, the location-based engine includes task assignment, resource simulation, monitoring, and reporting to achieve self-configuration for operations. Task assignment automatically dispatches tasks from the SL to the corresponding front-end operators immediately and reverts failed tasks to the SL for rematching of resources. Resource simulation provides simple comparisons between available and required resources to balance resources in different spaces. Monitoring and reporting modules handle the task dispatching results, resource simulation suggestions, and other execution results from the SL, for real-time feedback to users for decision-making.

5. Case study

5.1. Introduction to the case

A demonstration case is studied in the spare parts management department of Company H in Hong Kong, one of the largest MRO service providers in terms of capacity. SPI in Company H handles spare parts up to 250,000 SKU (stock keeping units) in 18 stores to serve 146 airlines with various aircraft types. SPI uses 48 delivery points for temporal storage near the maintenance sites to support multiple maintenance jobs. SPI has managed 700 parts receipts, 5,000 picking demands, and 4,000 delivery demands on average per day over the last seven years, with additional quality control and security check activities. SPI has a team of 300 workers and 20 vehicles to manage the SPI process. Company H uses a desktop portal system developed in 2008 for spare parts inventory management, with inaccurate and lagging SPI data. Since 2017, MRO providers have designed several new integrated platforms including AVIATAR, AARIVE, SMART MRO, and AMOS to integrate AI and big data technologies to support MRO activities and the spare parts supply chain [44–47]. Both the old portal system and the new integrated platforms provide spare parts inventory management, parts procurement, and human resource management functions to support SPI. However, these systems do not support SPI daily operations or the tracking of other resources related to SPI such as logistics units and vehicles. MRO providers must customize their internal SPI subsystems. As one of the largest MRO providers, Company H was innovative in introducing CPSPIS, which is complementary to integrated MRO platforms, integrating the SPI process and resources for SPI operation management in MRO industries.

5.2. Implementation and working logic

Our research team conducted the business process analysis through several site visits and interviews over one and a half years. The first step was identifying the participants and analyzing the process from a high level so that CPSPIS can systematically guide operations. Fig. 7 shows the complex and frequent collaboration of participants in SPI. The seven participants were categorized as requesters, end-users, and SPI working teams. After the high-level process analysis, all SPI activities were clarified. Table 1 defines the investigated requests, tasks, and resources according to the concepts in Section 3.2. All activities were consolidated into 10 requests and 26 tasks for the demonstration case.

Following the formulation in Section 3, we propose a technical entity design for CPSPIS, as shown in Fig. 8. According to the system architecture and the definitions, CPSPIS was developed using Java programming language in a Java runtime environment (JRE) 8.0. Tomcat 7.0 was used as the web application server and SQL Server 2012 was used as the database server, according to the company's requirements. A mobile application was used with Android 11.0.

Fig. 9 illustrates the implementation of CPSPIS. The SPI environment is transformed by intelligent IoT technologies. Beacons, smart sensors and light walls are deployed at delivery points, trolleys and material boxes respectively. These environmental data are visualized in resource management for monitoring and analysis. In addition, front-end operators are equipped with mobile industrial devices with warehousing apps and logistics apps installed to handle spare parts. The implementation of IoT technologies changes operator's daily operations into mobile and agile one and collects the operational data accurately. Finally, supervisors and managers monitor and trace real-time operation and resource data through the traceability and visibility tool for decision-making.

In addition to the working logic, Fig. 10 uses an example, 'Part Issuing Request', from the case scenario to illustrate how the services in Section 4 work. OFCS, HDSIS, ITTS, ESTIS and ESTES are developed into the corresponding automatic schedulers. A 'Parts Issuing Request' is created by dragging and attaching the related tasks on the 'Request Creation' page to initialize HDSIS. By clicking the request and task nodes, their properties are configured on the 'Request Definition' page. After saving the configuration, HDSIS is set up using these configured parameters.

Once HDSIS receives a part issuing request, it generates the request entity. Simultaneously OFCS and ITTS keep updating real-time available parts, workers, and logistics units on the 'Part List', 'User List', and 'Logistics Tool List'. ESTIS triggers a 'Part Picking' task with the available spare parts, worker and trolley by integrating the three resource lists and the 'Parts Issuing Request' entity. Otherwise, ESTIS triggers an 'Exception Handling' task for the request if there are no available parts.

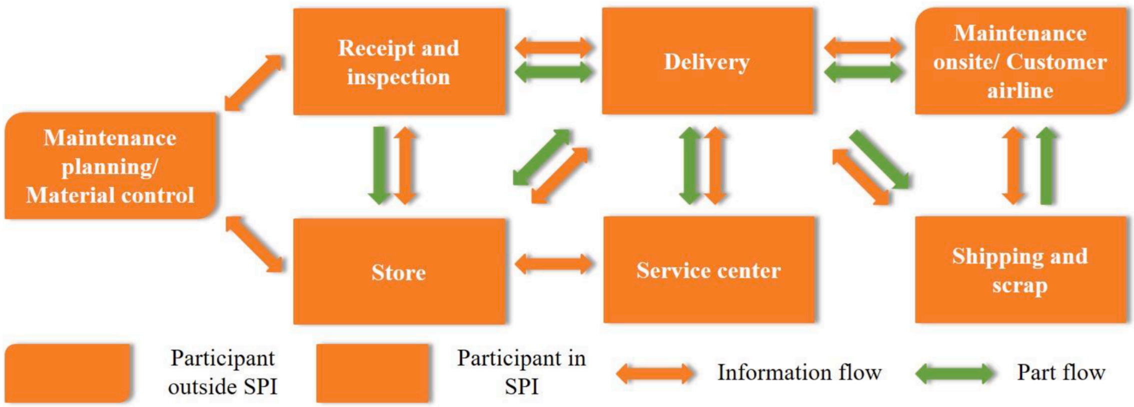

Fig. 7. SPI process analysis.

Table 1
Task, request and resource definition for CPSPIS.

Definition	Type
Task	order receipt, pallet receipt, part receipt, part inspection, quarantine handling, part delivery, part put-away, part labeling, part picking, picking sorting, picking grouping, part packing, delivery sorting, delivery grouping, part handover, part return, part scrap, air packing, stock counting, audit check, stock movement, additional job, data entry, exception handling, customs clearance, approval signature
Request	inbound handling, part issuing, U/S part return, serviceable part return, part cancellation, part scrap, storeToStore movement, stock take, audit check, air transport
Resource	trolley, material box, vehicle, mobile shelf, worker

ITTS processes ‘Part Sorting’ and ‘Part Grouping’ and assigns the task to a front-end operator on the ‘Picking Task List’ module of the warehousing application.

The assigned operator executes the task by physically obtaining the assigned trolley and spare parts of the ‘Part Picking’ task and recording the information in the mobile application system. Once ESTES receives the completed task, it triggers the ‘Part Labeling’ task using the real-time ‘User List’ and ‘Equipment List’ from OFCS and ITTS and the configuration from the ‘Part Issuing Request’. ITTS dispatches the task to the assigned operator for execution. During the progress above, ESTES keeps updating a workflow for tracing in the ‘Workflow List’.

5.3. Improvement and discussion

The proposed system addresses the synchronization problems in the process and the resources to improve SPI efficiency. The qualitative improvement of CPSPIS is analyzed from three perspectives. First, CPSPIS improves the collaborative efficiency of all participants through monitoring and tracing of the entire process. At the operational level, IoT-enabled resources quickly make the operators aware of appropriate equipment to complete their tasks. The interaction between the web system and mobile applications makes information transfer between operators, supervisors, and managers accurate and prompt. Moreover, CPSPIS reduces the communication time between the SPI participants by releasing them from switching between ERP systems, making phone calls, and sending emails. Second, CPSPIS optimizes the SPI process. CPSPIS consolidates all requests and activities into unified definitions (request, task, and workflow) to reduce duplicate operation steps. The organized process and semi-automatic tasks provide explicit instructions for operators to prevent missing activities. Early warning mechanisms and validation of the mobile applications reduce manual errors during execution. ESTIS and ESTES provide a spatial–temporal synchronization mechanism between the process and resource in real time to reduce the resource allocation time. Third, CPSPIS enhances decision support

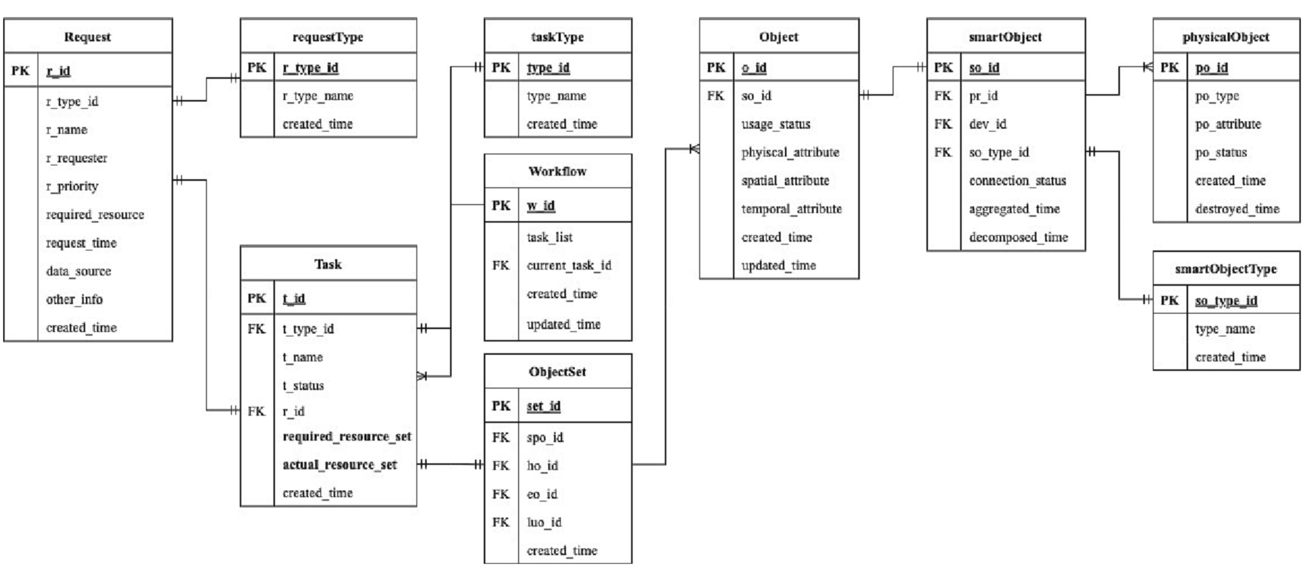

Fig. 8. Entity design for CPSPIS.

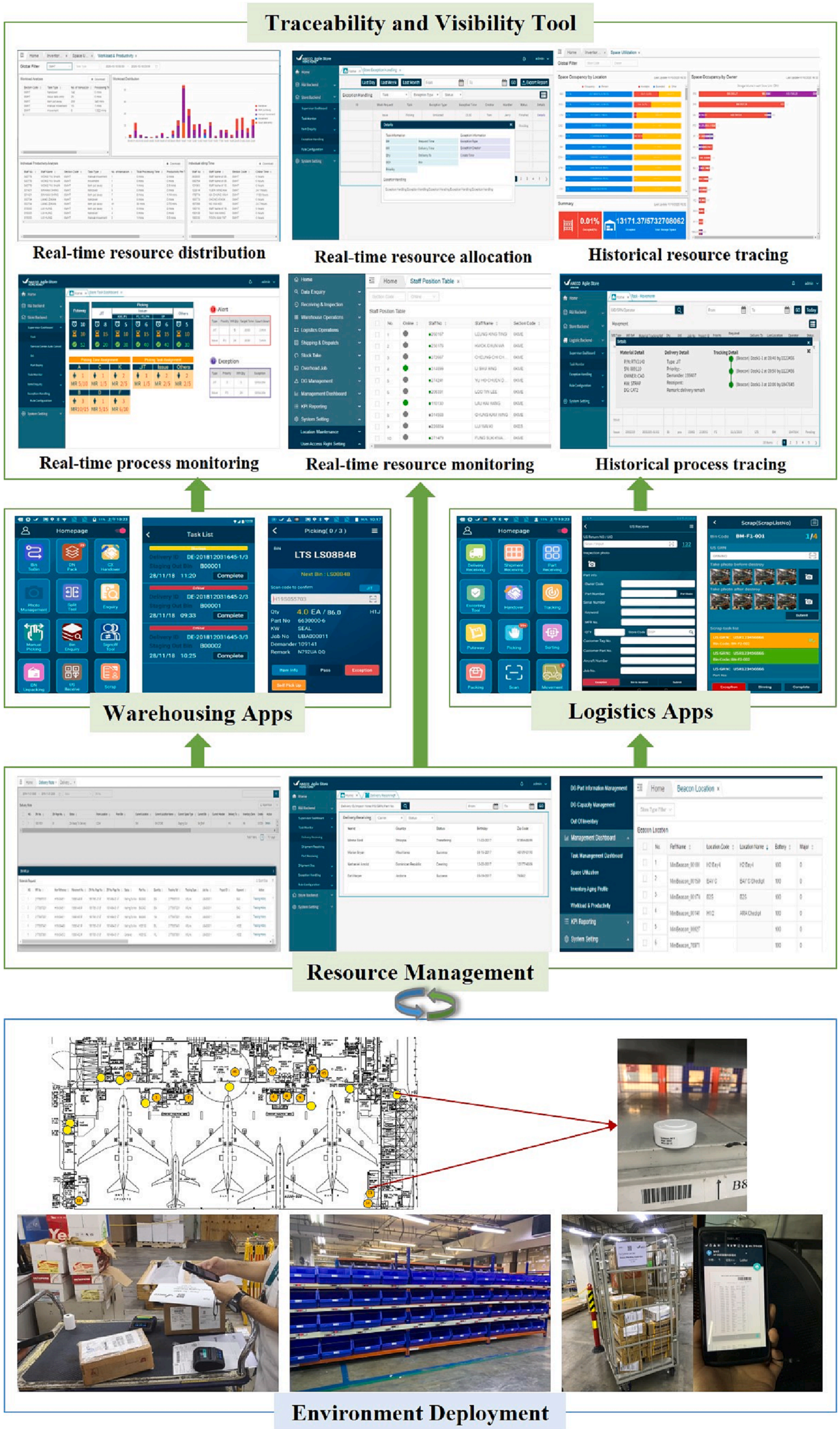

Fig. 9. Implementation of CPSPIS.

capability. The configuration of request, task, and workflow changes the operation decision from personal experience into knowledge-based rules to reduce uncertainty and inconsistency. The traceability and visibility tools provide real-time monitoring and analysis of resource and process data for managers. The quantitative improvement in terms of time and cost is summarized in Table 2.

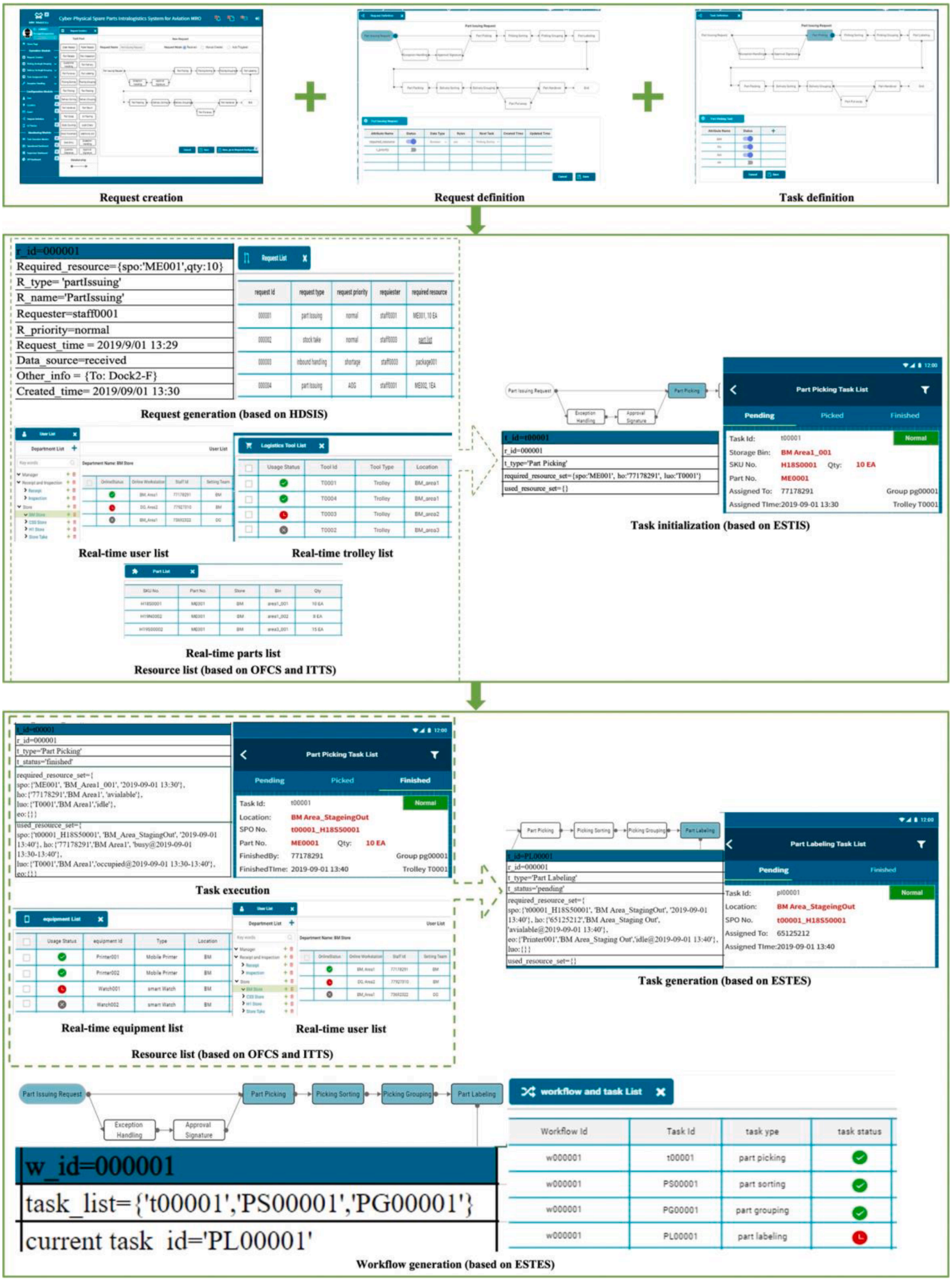

Fig. 10. An example illustrating the use of CPSPIS.

6. Conclusion

This study proposes CPSPIS for practical resource and operation management for SPI in aviation MRO. CPSPIS is designed as a five-layer structure with model-driven services for simple implementation. The overall framework of CPSPIS is presented, followed by the abstraction and encapsulation of physical resources and standardization of the business process. The design ends with business-related execution

Table 2
Quantitative improvement of CPSPIS.

KPI	Definition	Before	After
Paperwork	Size of label attached to spare part during each operation step for tracking purposes	297 mm* 210 mm /step Reduced 84.6% per step	60 mm*160 mm /step
Response time for environmental monitoring	Time required for recording temperature and humidity of storage environment	24 h Reduced 91.7%	1 h
Response time for task sorting and grouping	Time required for scheduling warehousing and logistics activities	1 d	10 min–1 h
Response time for inventory report	Time required for generating customer inventory report	24 h	≤ 1 h
Response time for resource allocation	Time required to obtain a worker or equipment to complete an activity	Random	1 min
Response time for collaboration	Time required for information transfer between two participants	Random	≤ 5 min
Response time for stock take	Time required for scheduling stock take activities	7 d	1 d
On-time picking in warehousing	Probability of picking correct spare parts from storage area by due time	Reduced 10% on average	
On-time delivery in logistics	Probability of delivering spare parts from store to destination by due time	Reduced 6% on average	
Manpower in warehouse team	Total manpower required to complete warehousing activities	Reduced 10%	
Manpower in logistics team	Total manpower required to complete logistics activities	Reduced 17%	

applications, a configurable workflow, and data visualization tools. Further, the intelligent mechanism of CPSPIS is illustrated using five self-X services. The effectiveness of CPSPIS is verified in a real-life case scenario.

The contributions of this paper are observed in three ways. First, CPSPIS realizes resource coordination and improves resource utilization. CPSPIS models all SPI resources as digital representations using IoT technologies to collect real-time spatial–temporal resource information. These resource information such as capacity and availability are traceable and visible, so that resource allocation can be adjusted in time during the execution of the SPI process. Second, CPSPIS facilitates the synchronization of SPI business processes for efficient management. The spare parts-oriented business process is methodically organized through the unified representations of task, request, and workflow. Initialization and execution services, with configurable workflows and flexible front-end operations, enhance timely cooperation among participants in SPI business processes. Third, CPSPIS synchronizes the business process and resources to enhance work efficiency at the execution, decision-making, and system levels. At the execution level, CPSPIS can immediately synchronize the front-end operations and simultaneously monitor the execution results through the designed applications to reduce execution and communication time. At the decision-making level, operational decisions are automatically performed by the self-X services, which can be flexibly adjusted based on accurate real-time resource and task information. At the system level, spare parts-oriented process monitoring, location-based process control, and whole-process SPI performance analysis are provided through plentiful user interfaces.

Implementation of CPSPIS in case company H indicates the possible benefits and improvements for SPI. However, CPSPIS can be further extended. After accurate real-time data is collected by CPSPIS in a real-life case, through data analysis, rule-based or semi-automatic decision-making can become automatic. In addition, resource prediction and operation optimization can be conducted using collected real-time data and history data after implementing CPSPIS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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